FACULTY OF COMPUTING

SEMESTER 2 2025/2026

SECP3213 BUSINESS INTELLIGENCE

SECTION 02

Project Phase 3

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1.0 Restaurant Performance Analysis Dashboard

1.1 Dashboard Overview

Dashboard was developed using Microsoft Power BI to evaluate restaurant sales performance based on the integrated dataset generated during the data preparation phase. The dashboard provides a comprehensive overview of key business performance indicators (KPIs), monthly revenue patterns, restaurant performance comparisons, and product sales performance. Interactive slicers were incorporated to enable users to filter and explore the data according to restaurant, season, month, and day of the week. The dashboard serves as a decision-support tool that assists management in monitoring operational performance and identifying opportunities for business improvement.

Figure 1.1 illustrates the Restaurant Performance Analysis Dashboard developed using Microsoft Power BI. The dashboard integrates multiple visualizations to provide a comprehensive overview of restaurant performance and support data-driven decision-making.

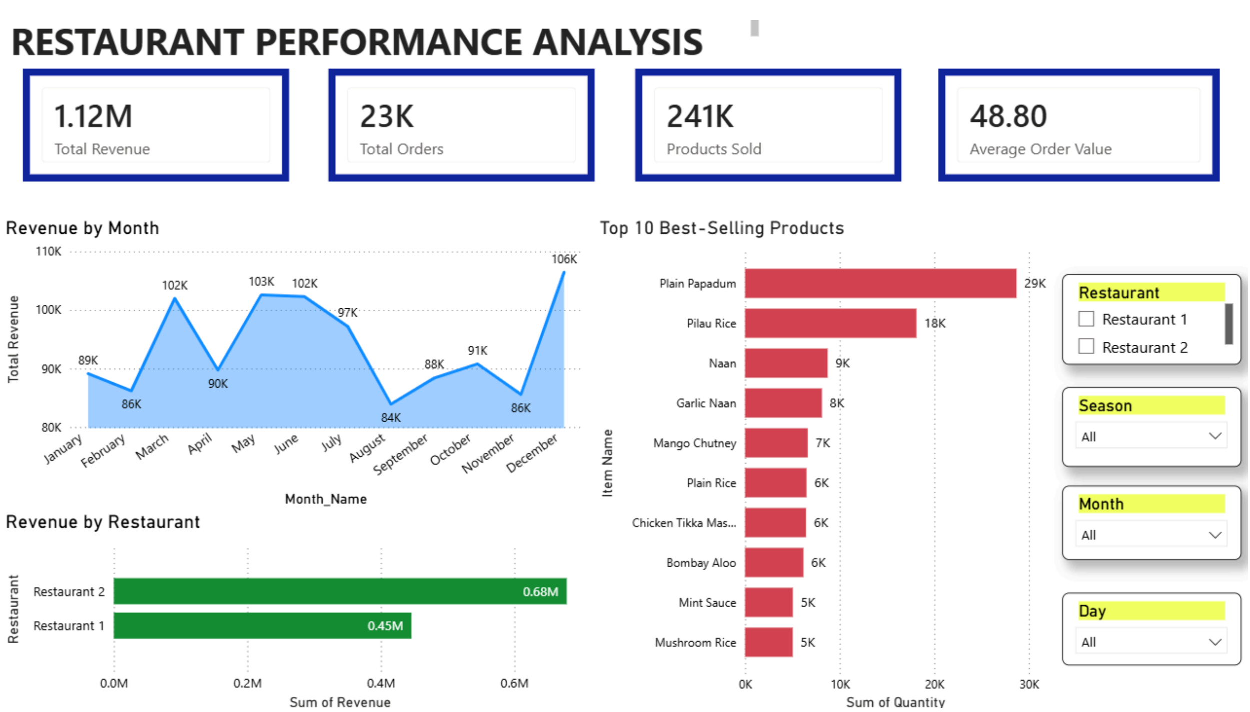

Figure 1.1: Restaurant Performance Analysis Dashboard

1.2 KPI Dashboard

The KPI section presents four key business metrics, namely Total Revenue, Total Orders, Products Sold, and Average Order Value. These indicators provide a summarized view of overall restaurant performance and allow stakeholders to evaluate business performance efficiently. Figure 1.2 illustrates the KPI Dashboard developed to provide a summary of the overall restaurant performance.

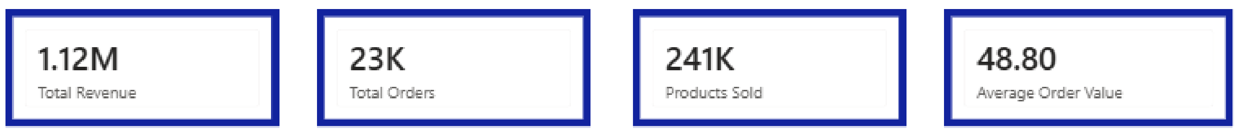

Figure 1.2: KPI Dashboard

KPI cards are commonly used in Business Intelligence dashboards to present important business metrics in a concise and easily interpretable format. The visualization enables decision-makers to quickly assess organizational performance without requiring detailed examination of transaction-level data.

Analysis of Results

The KPI dashboard indicates that the restaurants generated approximately £1.12 million in total revenue from more than 23,000 customer orders. In addition, approximately 241,000 products were sold during the analysis period. The average order value was £48.80 per transaction. These findings demonstrate strong sales activity and provide an overall assessment of restaurant business performance.

1.3 Revenue by Month

A line chart was used to visualize revenue performance across different months. The chart illustrates revenue fluctuations throughout the year and highlights periods of strong and weak sales performance.Figure 1.3 illustrates the monthly revenue analysis used to evaluate revenue performance across different months.

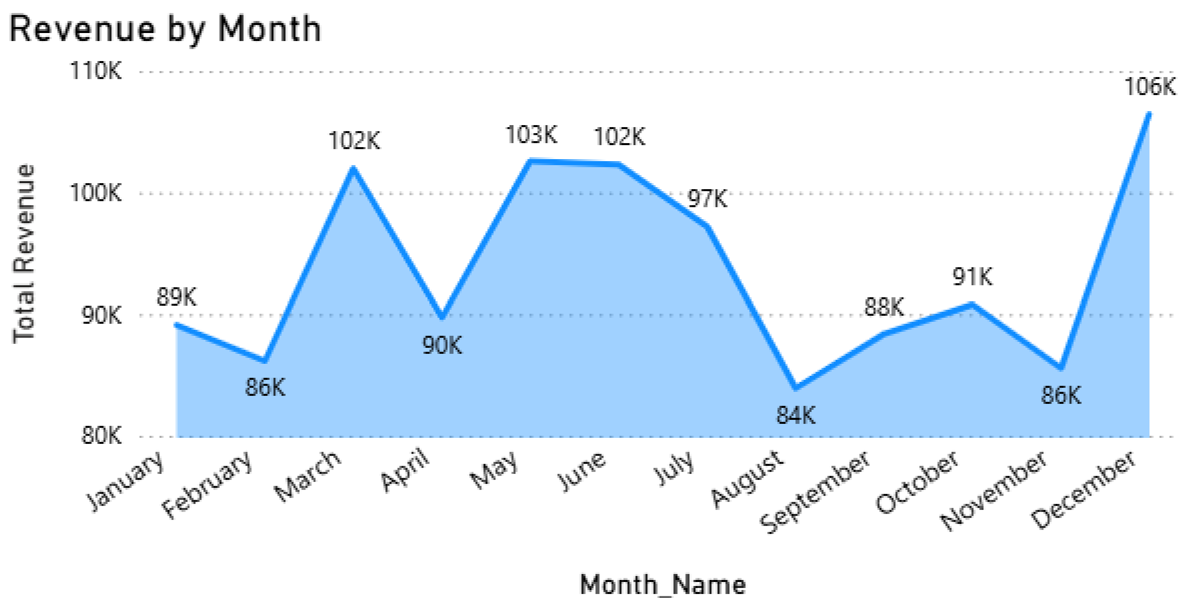

Figure 1.3: Revenue by Month

Line charts are highly suitable for trend analysis because they effectively display changes in performance over time. This visualization enables management to identify seasonal patterns, monitor sales fluctuations, and evaluate business performance across different periods.

Analysis of Results

The monthly revenue analysis revealed that December recorded the highest revenue, reaching approximately £106,000. In contrast, August generated the lowest revenue of approximately £84,000. Although revenue fluctuated throughout the year, the overall trend suggests relatively stable sales performance. The stronger performance observed in December may be attributed to increased consumer spending during year-end and festive periods.

1.4 Revenue by Restaurant

A horizontal bar chart was used to compare the total revenue generated by Restaurant 1 and Restaurant 2. The chart provides a direct comparison of the sales performance achieved by each restaurant.Figure 1.4 illustrates the comparison of revenue generated by Restaurant 1 and Restaurant 2.

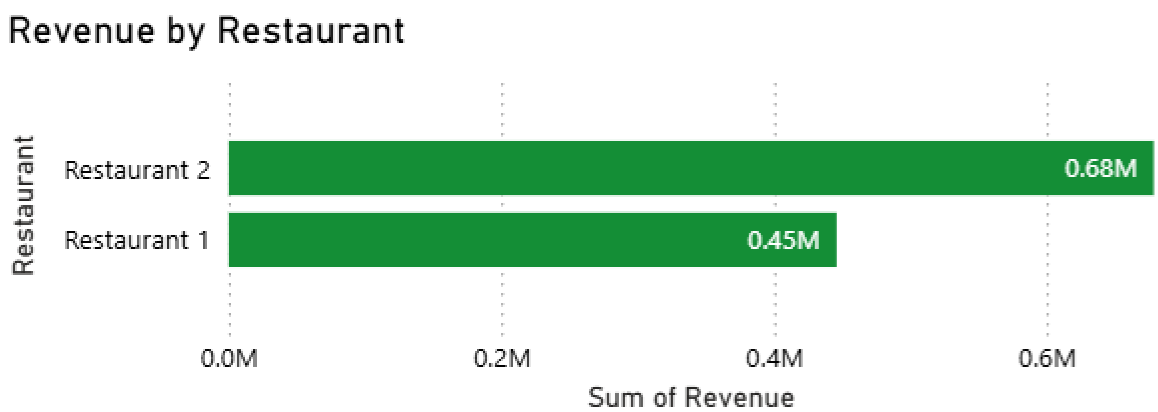

Figure 1.4: Revenue by Restaurant

Bar charts are effective for comparing performance across categories because differences in values can be identified quickly and accurately. This visualization assists management in evaluating the relative contribution of each restaurant to overall revenue generation.

Analysis of Results

The analysis indicates that Restaurant 2 generated approximately £0.68 million in revenue, whereas Restaurant 1 generated approximately £0.45 million. The findings suggest that Restaurant 2 contributed a substantially larger share of total revenue and demonstrated stronger overall sales performance during the analysis period.

1.5 Top 10 Best-Selling Products

A horizontal bar chart was used to display the ten products with the highest sales volume. The chart ranks products according to quantity sold and highlights customer purchasing preferences. Figure 1.5 illustrates the top ten best-selling products based on sales volume.

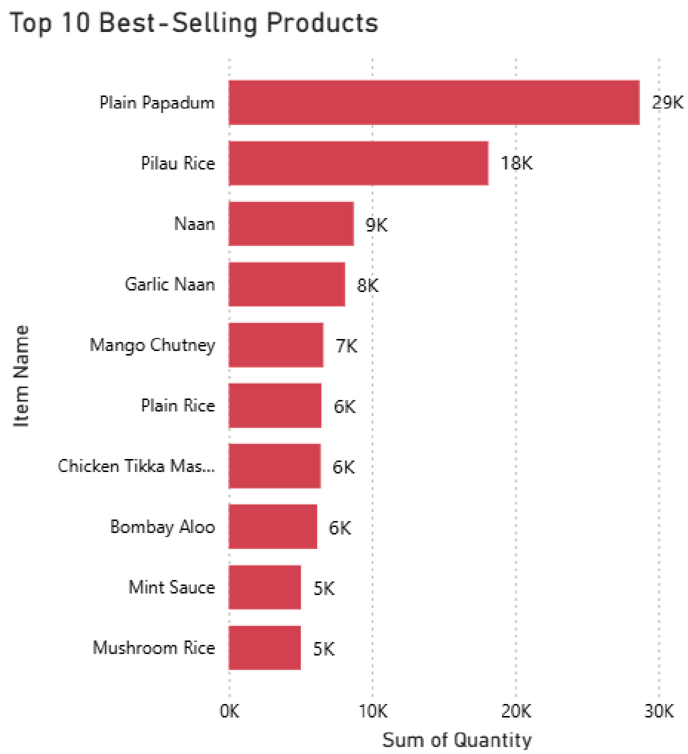

Figure 1.5: Top 10 Best-Selling Products

This visualization is appropriate for identifying product popularity and evaluating product performance. By ranking products according to sales volume, management can identify high-demand items and make informed inventory and marketing decisions.

Analysis of Results

The product sales analysis revealed that Plain Papadum was the highest-selling menu item, achieving approximately 29,000 units sold. Pilau Rice ranked second with approximately 18,000 units sold, followed by Naan with approximately 9,000 units sold. These findings indicate that a small number of menu items contribute significantly to overall product sales and represent important revenue-generating products for the restaurants.

2.0 Weather Impact Analysis Dashboard

2.1 Dashboard Overview

The Weather Impact Analysis Dashboard was developed using Microsoft Power BI to evaluate the influence of weather conditions on restaurant sales performance and customer demand. The dashboard integrates weather attributes and sales data to provide insights into the relationship between temperature, precipitation, seasonal patterns, and restaurant revenue. Interactive slicers were incorporated to enable users to filter and explore the data according to restaurant, season, and weather category. The dashboard serves as a decision-support tool that assists management in understanding how weather conditions influence business performance and customer purchasing behaviour.

Figure 2.1 illustrates the Weather Impact Analysis Dashboard developed using Microsoft Power BI. The dashboard integrates multiple visualizations to analyze weather-related factors and their impact on restaurant sales and customer demand.

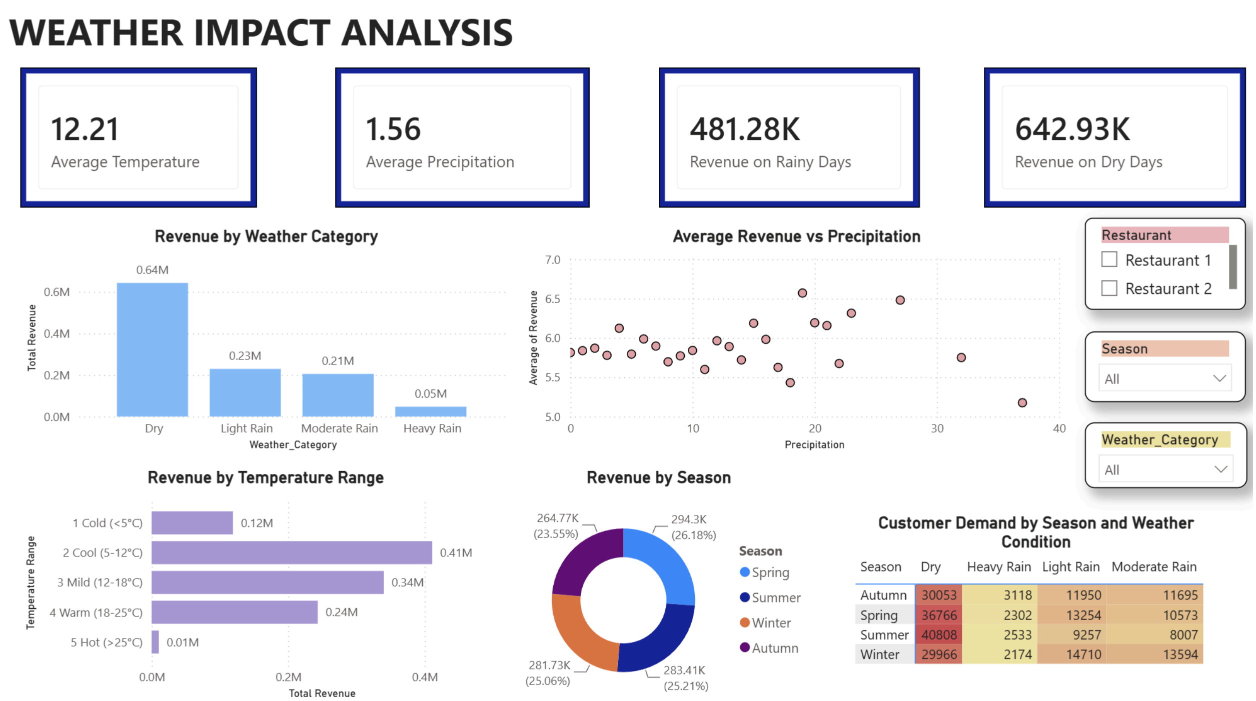

Figure 2.1: Weather Impact Analysis Dashboard

2.2 KPI Dashboard

The KPI section presents four weather-related business metrics, namely Average Temperature, Average Precipitation, Revenue on Rainy Days, and Revenue on Dry Days. These indicators provide a summarized overview of weather conditions and their corresponding impact on restaurant revenue. Figure 2.2 illustrates the KPI Dashboard developed to provide a summary of weather-related business performance.

Figure 2.2: KPI Dashboard

KPI cards are widely used in Business Intelligence dashboards to summarize important metrics in a concise and easily understandable format. The visualization enables decision-makers to evaluate weather conditions and assess their influence on business performance without analysing detailed transaction records.

Analysis of Results

The KPI dashboard indicates that the average temperature recorded during the analysis period was approximately 12.21°C, while the average precipitation level was 1.56 mm. Revenue generated during rainy weather conditions amounted to approximately £481,280, whereas revenue generated during dry weather conditions reached approximately £642,930. These findings suggest that dry weather contributed more significantly to restaurant revenue compared to rainy weather conditions.

2.3 Revenue by Weather Category

A stacked column chart was used to compare total revenue generated under different weather categories, namely Dry, Light Rain, Moderate Rain, and Heavy Rain. The chart provides a clear comparison of sales performance across various weather conditions. Figure 2.3 illustrates the revenue distribution across different weather categories.

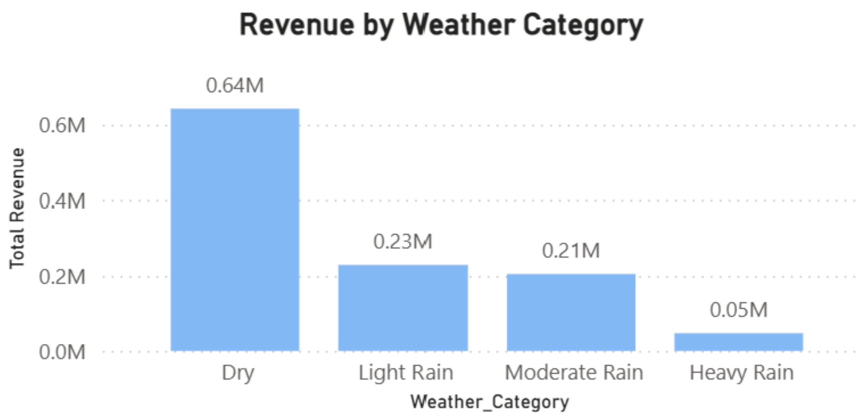

Figure 2.3: Revenue by Weather Category

Column charts are effective for comparing numerical values across multiple categories. This visualization enables management to identify weather conditions associated with stronger or weaker sales performance.

Analysis of Results

The analysis revealed that Dry weather generated the highest revenue, reaching approximately £640,000. Light Rain and Moderate Rain contributed approximately £230,000 and £210,000, respectively. Heavy Rain recorded the lowest revenue of approximately £50,000. These findings suggest that favourable weather conditions encourage greater customer purchasing activity, while heavy rainfall may negatively impact restaurant sales.

2.4 Average Revenue vs Precipitation

A scatter chart was used to analyse the relationship between precipitation levels and average revenue. The chart visualizes how revenue changes across different rainfall intensities and assists in identifying potential correlations between weather conditions and sales performance. Figure 2.4 illustrates the relationship between precipitation and average revenue.

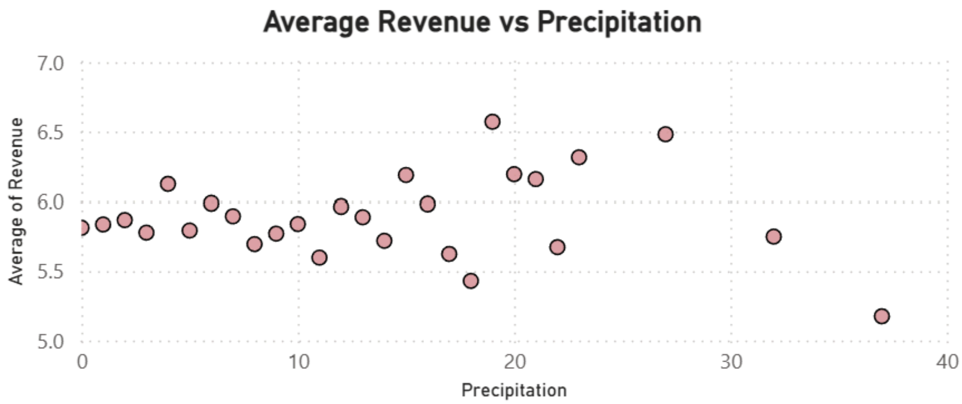

Figure 2.4: Average Revenue vs Precipitation

Scatter charts are appropriate for analysing relationships between numerical variables because they enable users to observe trends, patterns, and potential correlations. This visualization assists management in understanding whether rainfall intensity influences restaurant revenue.

Analysis of Results

The scatter plot indicates that average revenue generally fluctuates between £5.50 and £6.50 across different precipitation levels. Although no strong linear relationship is observed, variations in revenue can be identified under different rainfall conditions. The findings suggest that precipitation may influence customer purchasing behaviour, although additional factors may also contribute to sales performance.

2.5 Revenue by Temperature Range

A stacked bar chart was used to compare revenue generated across different temperature ranges, namely Cold, Cool, Mild, Warm, and Hot conditions. The chart illustrates how temperature variations affect restaurant revenue. Figure 2.5 presents the revenue generated under different temperature ranges.

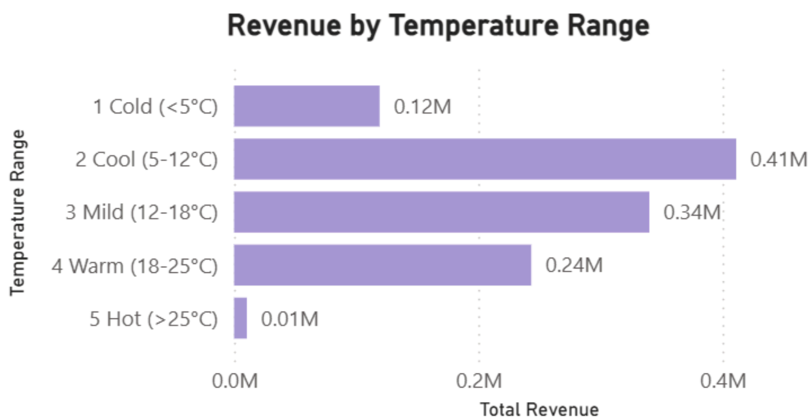

Figure 2.5: Revenue by Temperature Range

Bar charts are suitable for comparing performance across categorical groups because differences in values can be easily identified. This visualization enables management to evaluate which temperature conditions are most favourable for business performance.

Analysis of Results

The analysis indicates that the Cool temperature range (5–12°C) generated the highest revenue of approximately £410,000. This was followed by Mild temperatures with approximately £340,000 and Warm temperatures with approximately £240,000. Cold temperatures generated approximately £120,000, while Hot temperatures recorded the lowest revenue at approximately £10,000. These findings suggest that moderate weather conditions are associated with stronger sales performance.

2.6 Revenue by Season

A donut chart was used to visualize the distribution of total revenue across four seasons, namely Spring, Summer, Autumn, and Winter. The chart highlights the relative contribution of each season to overall revenue generation. Figure 2.6 illustrates the seasonal revenue distribution.

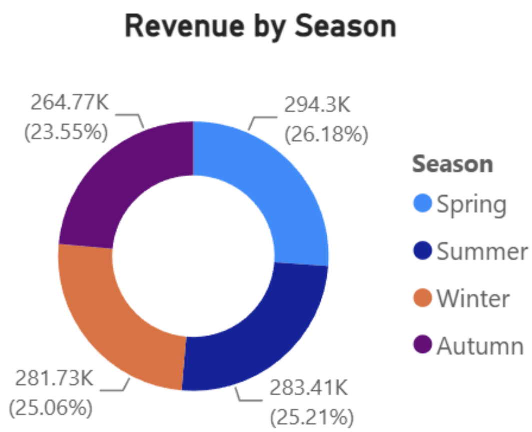

Figure 2.6: Revenue by Season

Donut charts are effective for illustrating proportional contributions among categories. This visualization enables management to compare seasonal performance and identify periods associated with stronger revenue generation.

Analysis of Results

The seasonal analysis indicates that revenue was relatively evenly distributed across all four seasons. Spring generated the highest revenue of approximately £294,300, followed by Summer with approximately £283,400, Winter with approximately £281,700, and Autumn with approximately £264,800. The findings suggest that restaurant sales remain relatively stable throughout the year, with only minor seasonal variations.

2.7 Customer Demand by Season and Weather Condition

A heatmap was used to analyse customer demand across different seasons and weather conditions. The visualization displays the quantity of products sold under each weather category and season combination. Figure 2.7 illustrates customer demand patterns under varying weather conditions.

Customer Demand by Season and Weather Condition

Season	Dry	Heavy Rain	Light Rain	Moderate Rain
Autumn	30053	3118	11950	11695
Spring	36766	2302	13254	10573
Summer	40808	2533	9257	8007
Winter	29966	2174	14710	13594

Figure 2.7: Customer Demand by Season and Weather Condition

Heatmaps are effective for identifying patterns and intensity differences across multiple dimensions. This visualization enables management to evaluate how weather conditions influence customer demand throughout the year.

Analysis of Results

The heatmap reveals that Dry weather consistently generated the highest customer demand across all seasons. Summer recorded the highest demand under Dry weather with approximately 40,808 products sold, followed by Spring with approximately 36,766 products sold. Heavy Rain consistently recorded the lowest demand across all seasons. These findings indicate that customer purchasing activity is generally stronger during favourable weather conditions and decreases during periods of heavy rainfall.

3.0 Storyboard

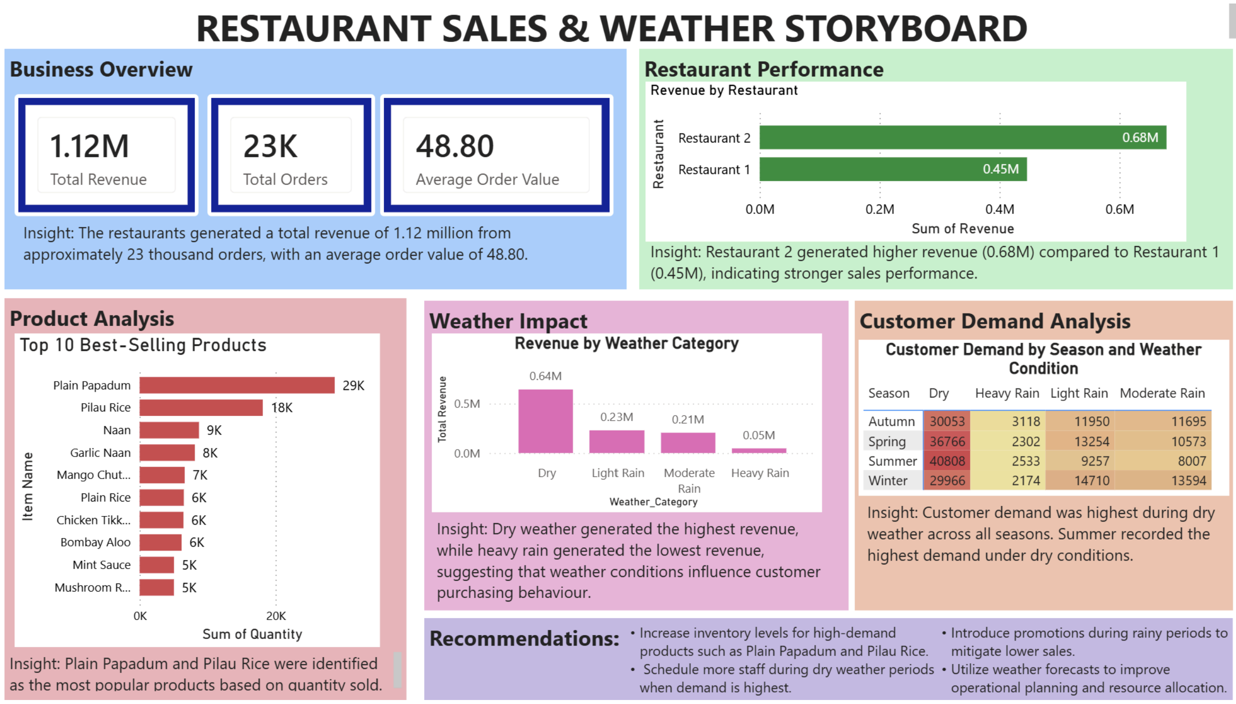

Figure 3.1: Restaurant Sales and Weather Storyboard

A storyboard was developed using Microsoft Power BI to summarize the key findings obtained from the Restaurant Performance Analysis Dashboard and Weather Impact Analysis Dashboard. The storyboard integrates important business metrics, sales performance indicators, product analysis, weather-related insights, customer demand patterns, and business recommendations into a single page. This allows stakeholders to quickly understand the overall business performance and identify important factors influencing restaurant sales.

Figure 3.1 illustrates the Restaurant Sales and Weather Storyboard developed to present the major findings generated from the dashboard analyses. The storyboard highlights that the restaurants generated approximately £1.12 million in revenue from more than 23,000 customer orders, indicating strong overall business performance. Restaurant 2 recorded higher revenue compared to Restaurant 1, while Plain Papadum and Pilau Rice were identified as the most popular menu items based on quantity sold.

The storyboard also demonstrates the impact of weather conditions on restaurant performance. Dry weather generated the highest revenue and customer demand, whereas heavy rain was associated with lower sales performance. In addition, customer demand remained highest during dry weather conditions across all seasons, particularly during Summer. Based on these findings, several recommendations were proposed, including maintaining adequate inventory for high-demand products, increasing staffing levels during peak demand periods, implementing promotional campaigns during rainy weather, and utilizing weather forecasts to support operational planning and decision-making.

4.0 Key Insights

4.1 Insights from Restaurant Performance Analysis Dashboard

1. The restaurants generated approximately £1.12 million in total revenue from more than 23,000 customer orders, indicating strong overall business performance.
2. Restaurant 2 outperformed Restaurant 1 by generating approximately £230,000 more revenue, demonstrating a stronger contribution to overall sales.
3. December recorded the highest monthly revenue, while August generated the lowest revenue, suggesting the existence of seasonal variations in customer demand.
4. Plain Papadum and Pilau Rice emerged as the most popular menu items, indicating strong customer preference for these products.

4.2 Insights from Weather Impact Analysis Dashboard

1. Dry weather generated the highest revenue of approximately £640,000, whereas Heavy Rain recorded the lowest revenue of approximately £50,000, indicating that weather conditions influence restaurant sales performance.
2. The Cool temperature range (5–12°C) generated the highest revenue of approximately £410,000, while Hot temperatures (>25°C) contributed the lowest revenue, suggesting that moderate temperatures are more favourable for customer purchasing activity.
3. Revenue remained relatively balanced across Spring, Summer, Autumn, and Winter, indicating that restaurant sales were not heavily dependent on seasonal changes.
4. Customer demand was consistently highest during Dry weather conditions across all seasons. Summer recorded the highest demand under Dry weather with approximately 40,808 products sold.
5. The scatter chart analysis showed fluctuations in average revenue across different precipitation levels, suggesting that rainfall may influence customer purchasing behaviour and sales performance.

5.0 Business Recommendations

5.1 Recommendations from Dashboard 1

1. Management should investigate the factors contributing to the superior performance of Restaurant 2 and consider implementing similar operational strategies in Restaurant 1.
2. Adequate inventory levels should be maintained for high-demand products such as Plain Papadum and Pilau Rice to prevent stock shortages and potential revenue loss.
3. Promotional campaigns should be intensified during lower-performing periods, particularly in August, to stimulate customer demand.

5.2 Recommendations from Dashboard 2

1. Restaurant management should anticipate increased customer demand during Dry weather conditions by ensuring sufficient staffing levels and inventory availability.
2. Promotional campaigns, discounts, and delivery incentives should be introduced during rainy weather periods to mitigate potential revenue losses.
3. Weather forecasts should be incorporated into operational planning to improve inventory management, workforce scheduling, and service preparation.
4. High-demand products should be prioritized during favourable weather conditions to maximize sales opportunities and customer satisfaction.
5. Continuous monitoring of weather-related sales patterns through the dashboard should be encouraged to support data-driven decision-making and improve business performance.